EQUITY RESEARCH WHITE PAPER
Costco Wholesale Corporation (NASDAQ: COST)

MBAN 5570 – Accounting and Financial Analytics

Dr. Mohammad Rahaman

Prepared by:
Pamela Ballesteros
Irene Amoakohene

1. Executive Summary & Investment Thesis

Costco Wholesale Corporation (NASDAQ: COST) is a best-in-class warehouse retailer operating on a membership-subsidized, ultra-low-margin merchandise model. Over the five-year period from FY2021 to FY2025, Costco compounded revenue at a 4-year CAGR of 6.6%, exceeding the 5% benchmark, and net income at an 11.5% CAGR, comfortably above the 7% benchmark. Free cash flow reached a record $7.84B in FY2025, up 18% year-over-year, with a FCF/Net Income conversion ratio of approximately 96%, signalling high earnings quality with minimal accrual risk.

At a current price of approximately $983, Costco trades at ~51× trailing earnings, 31% above its 10-year historical average P/E of 39× and 2–4× the multiples commanded by retail peers. Our base-case Discounted Cash Flow (DCF) analysis yields an intrinsic value of $850–$950, implying the stock is fairly valued to modestly overvalued at current levels. Only under optimistic assumptions (FCF growth ≥15%, WACC ≤8.5%) does the model justify today's market price.

Our investment rating is HOLD for existing shareholders, with a recommendation to Accumulate on Weakness below $900. The quality of the business is exceptional and not in question. What is in question is whether current market pricing already reflects near-perfection.

2. Business Model & Competitive Advantage

Costco's economic moat rests on four interlocking and self-reinforcing pillars that are structurally near impossible to replicate at scale.

Membership Flywheel

The membership model is the financial bedrock of the entire business. Annual fees generate approximately $7B in near-pure-margin income, funding the ability to price merchandise at deliberately thin margins (branded SKUs capped at ~14% markup; Kirkland Signature at ~15%). The flywheel runs as follows: members pay the fee → Costco caps merchandise margins → members perceive they 'earn back' the fee → 92.1% of US members renew → Costco collects more fee income → prices fall further. This self-reinforcing loop has operated for over 40 years and grows more durable each cycle.

Low-SKU Strategy & Supplier Power

With only ~4,000 active SKUs versus 30,000+ at Walmart, Costco concentrates purchasing volume per product to a degree that gives it substantial pricing power over suppliers. At $269.9B in purchasing volume, Costco largely dictates supplier terms, positioning it favourably on Porter's Five Forces (Supplier Bargaining Power: LOW).

Kirkland Signature

Contributing approximately 30% of total revenue at higher-than-branded margins, Kirkland Signature is Costco's most defensible single asset against digital disruption. Unlike commodity merchandise that can be replicated on Amazon, Kirkland represents trusted quality at below-market prices, a perception that took decades to build and cannot be quickly disintermediated.

International Growth & E-Commerce

International operations generate ~34% of operating income from only ~27% of sales, implying superior per-unit economics. Markets including France, Spain, and China represent multi-decade expansion runways. Digital comparable sales grew +20.9% in Q2 FY2025 and +15.6% for the full year, though penetration relative to the total member base remains low, a meaningful growth optionality embedded in the current valuation.

3. Corporate Governance & Capital Allocation

Governance Metric	Detail
Institutional Ownership	~68%
Insider Ownership	~0.23% (low, aligned via comp structure)
Accounting Restatements	None on record
CEO Compensation Link	Tied to membership growth and comp sales, incentives aligned with long-run value creation
Dividend History	20+ consecutive annual increases; 5 special dividends since 2013
Audit / Board Risk Score	Low (1/10 on internal risk scorecard)

Special Dividend (Jan 2024)	$15.00/share, next cycle estimated $15–20/share (2026–2027)
Share Buybacks (FY2025)	$903M, disciplined, not value-destructive at these multiples
ROIC (FY2025)	37.8%, materially above WACC (~9%), strong value creation

Management's track record of capital allocation is one of the strongest in large-cap retail. The lack of accounting restatements and the consistent linkage of executive compensation to operational metrics rather than short-term earnings manipulation are meaningful positives. The slight insider ownership figure (0.23%) is a standard characteristic of large-cap consumer staples and is offset by the incentive structure.

4. Industry Analysis & Critical Risk Factors

4.1 Industry Attractiveness

The warehouse club segment is structurally attractive relative to broader retail: barriers to entry are high (Costco's 46% US market share reflects 25+ years of capital investment, member trust, and operational discipline), and the membership model insulates revenue from short-term consumer spending volatility. Porter's Five Forces analysis reveals two LOW-force categories (new entrants, supplier power) against three MEDIUM-force categories (buyer power, substitutes, rivalry). This is an above-average industry structure, and Costco commands it from the dominant market position.

4.2 Critical Risk Factors

- Valuation Risk (High): At 51× trailing P/E, 31% above the 10-year average, any earnings miss or sentiment shift could trigger meaningful multiple compression. Reverting to the 10-year average P/E alone implies approximately 24% downside with no change in earnings.
- Tariff & Supply Chain Risk (Medium-High): CFO commentary in Q2 FY2025 flagged that grocery margins are 'much tighter' due to import tariff escalation. Further escalation on China-sourced goods would impair the Kirkland Signature supply chain and structurally threaten Costco's core low-price promise.
- Membership Renewal Erosion (Medium-High): US renewal rate declined marginally from 93% to 92.1%. The entire financial model is subsidized by fee income. A sustained decline below 90% would fundamentally impair the operating leverage underpinning the investment thesis.
- E-Commerce & Amazon Rivalry (Medium): Amazon Prime + Whole Foods continues converging on Costco's value proposition, particularly for urban members. Digital penetration growth at Costco offsets this partially, but it remains a structural long-run threat.
- Interest Rate / WACC Shock (Black Swan): Monte Carlo variance attribution identifies WACC as the single largest driver of intrinsic value (66.7% of variance). A WACC increase from 9% to 11% reduces base-case intrinsic value by approximately $130/share, pushing the stock into materially overvalued territory under nearly all scenarios.

5. Financial Statement Analysis

5.1 Income Statement, FCF & Capital Efficiency (FY2021–FY2025)

Metric	FY2021	FY2022	FY2023	FY2024	FY2025
Revenue ($B)	$192.1	$222.7	$242.3	$249.6	$269.9
Net Income ($B)	$5.01	$5.84	$6.29	$7.37	~$8.0
EPS	$11.27	$13.14	$14.16	$16.56	~$18.05
FCF ($B)	$5.37	$3.50	$6.75	$6.63	$7.84
Gross Margin	12.5%	12.3%	12.6%	12.7%	12.84%
Net Margin	2.61%	2.62%	2.60%	2.95%	2.97%
ROIC	—	~30%	~33%	~36%	37.8%

5.2 Key Observations

- Revenue CAGR (4-year): 6.6%, exceeds the 5% benchmark, supported by new warehouse openings, e-commerce growth, and the September 2024 membership fee increase.
- Net Income CAGR (4-year): 11.5%, comfortably exceeds the 7% benchmark. Driven by SG&A leverage, membership fee scaling, and Kirkland Signature margin accretion.
- FCF Quality: The 5-year average FCF/Net Income ratio of ~96% confirms low accrual content in earnings. FCF is real cash, not an accounting artefact. FY2022's dip to $3.50B was driven by elevated capex for new warehouses, not earnings deterioration.
- Margins: Deliberately thin gross margins (12.84% in FY2025) are a competitive feature, not a weakness. The margin structure is intentional, lower merchandise margins drive member retention, and membership fees carry near-100% gross margin. All three margin lines expanded from FY2020 to FY2025.
- Balance Sheet: ROIC of 37.8%, materially above WACC (~9%), confirms consistent, high-quality value creation. The balance sheet carries manageable leverage, operating cash flow of $14.76B (TTM, approximately 3× FY2020 levels), and a cash reserve of ~$17.38B.
- Capital Intensity: Capex is moderate (warehouses have long useful lives), and maintenance capital requirements are low relative to free cash flow generation. The business does not require heavy reinvestment to sustain competitive position, a hallmark of high-quality compounders.

6. Valuation & Forward Guidance

6.1 Multiples vs. Historical Averages

Metric	Current	5-Yr Avg	10-Yr Avg	Signal
Trailing P/E	~51×	45×	39×	⚠ Stretched +31%
Forward P/E	~46×	41×	35×	⚠ Stretched +31%
EV/EBITDA	31.4×	~27×	~23×	⚠ Elevated +37%
P/FCF	~50×	~48×	~38×	⚠ Elevated +32%
Dividend Yield	~0.5%	~0.8%	~1.1%	Below historical avg

6.2 DCF Scenario Analysis

A Monte Carlo DCF simulation was conducted (see Exhibit A) which our model applies a 5-year explicit FCF projection period followed by a Gordon Growth terminal value. Base FCF of $7.84B (FY2025). Shares outstanding: ~444M. A 10% margin of safety is applied throughout.

Scenario	Intrinsic Value Range
Bear Case (6% FCF growth, WACC 10.5%)	$650–$750
Base Case (10% FCF growth, WACC 9.0%)	$850–$950
Bull Case (15% FCF growth, WACC 8.5%)	$1,100–$1,200
Analyst Consensus Target (40+ analysts)	~$1,088 (implied +10.7% upside)
Current Market Price (April 2026)	~$983, requires bull-case assumptions

Only at FCF growth ≥12% with WACC ≤9%, or FCF ≥15% with WACC ≤10%, does the DCF model justify today's market price. The gap between our base-case intrinsic value ($850–$950) and the analyst consensus target ($1,088) is best explained by the persistent quality premium the market assigns to Costco's membership moat and earnings predictability, a premium that is historically justified but which limits near-term upside from current levels.

6.3 Peer Comparison

Company	Fwd P/E	EV/EBITDA	Rev Growth	Moat
COST (Costco)	**~46×**	**31.4×**	**~8%**	**Membership + scale**
WMT (Walmart)	~38×	~22×	~5%	Scale + omnichannel
BJ's Wholesale	~19×	~12×	~3%	Regional membership
TGT (Target)	~12×	~8×	~1%	Brand + private label
Sector Median	~21×	~14×	~4%	—

Costco commands a 2–4× valuation premium relative to retail peers. This premium is fundamentally justified by the membership moat, superior ROIC, and FCF quality, but it creates a high bar for new entrants at current pricing.

6.4 Shareholder Value Creation

Shareholders' earnings = EPS growth + Dividend yield. Over the past four fiscal years, Costco compounded EPS from $11.27 (FY2021) to ~$18.05 (FY2025), representing a CAGR of approximately 12.5%, comfortably exceeding the 10% benchmark. Adding the ~0.5% regular dividend yield and accounting for special dividends (five payments since 2013, including the $15.00/share in January 2024), total shareholder returns have significantly exceeded the 10% threshold. The September 2024 membership fee increase (Executive tier: $120→$130) is expected to contribute approximately $0.35–$0.40 to EPS in FY2026, providing a near-term earnings catalyst. Based on our analysis, sustaining >10% compounding in shareholders' earnings over the next 3–5 years is achievable but requires the membership renewal rate to stabilize above 92% and tariff headwinds to remain manageable.

6.5 Forward Guidance

TTM revenue through February 2026 reached $286.3B (+8.4% YoY), and TTM net income of $8.55B implies continuation of the 7–9% CAGR trajectory. Management has guided to 25–30 net new warehouse openings annually, with international stores (generating ~34% of operating income from ~27% of sales) representing the most accretive component of growth. A special dividend in 2026 or early 2027 is likely given operating cash flow of $14.76B and the historical 2–3-year special dividend cadence. E-commerce growth sustaining above 15% annually represents meaningful upside optionality not fully priced into our base case.

SECTION 2B & 2C: AI-ASSISTED ANALYSIS & CRITICAL EVALUATION

7. AI-Assisted Financial Analysis (Section 2B)

We employed two AI-assisted tools as analytical complements to the traditional analyses above: (1) Gemini (Google's large language model) to generate a summary equity research narrative; and (2) a Monte Carlo simulation (10,000 iterations) to model intrinsic value distributions across stochastic inputs.

7.1 Gemini-Generated Analysis, Summary of Outputs

- Executive Summary: Gemini characterized Costco's 2022–2026 period as one of 'stable, high-margin expansion' and correctly identified membership renewal rates >90% and predictable fee income as core earnings drivers.
- Revenue & Earnings Figures: Gemini cited TTM revenue of $286.27B, net income of $8.55B for FY2026 (estimated), and a 7–9% CAGR, figures broadly consistent with SEC-reported data and our independent research.
- Industry Trends: Gemini noted e-commerce growth of ~22% in the prior year and referenced AI-driven checkout automation increasing warehouse throughput by ~20% in 2025/2026.
- Risk Assessment: Gemini identified the P/E premium (~52×) as a valuation risk, geopolitical supply chain exposure, and tariff uncertainty, all consistent with our human-led risk register.
- Monte Carlo Simulation: The model sampled FCF growth (μ=10%, σ=3%), WACC (μ=9%, σ=1.2%), and terminal growth rate (μ=3%, σ=0.5%) across 10,000 iterations, generating a distribution of intrinsic value estimates with WACC identified as the dominant driver of variance (66.7%).

8. Critical Evaluation of AI Outputs (Section 2C)

8.1 What AI Got Right

Category	Detail
Membership Moat	Gemini correctly identified the membership renewal rate (>90%) as the structural anchor of the business model and correctly linked fee income to near-100% gross margin, consistent with our Porter's Five Forces analysis.
Revenue & Earnings Trajectory	The figures cited ($286.27B TTM revenue, $8.55B net income, 7–9% CAGR) are broadly consistent with SEC filings and our independent verification. AI captured the directional trend accurately.
Valuation Risk at 52× P/E	Gemini flagged the valuation premium over peers (P/E ~52× vs. Walmart ~28×) as the primary risk factor. This is directionally correct and aligned with our multiples analysis showing COST at 31% above its 10-year average P/E.
Monte Carlo Variance Attribution	The simulation's finding that WACC drives 66.7% of intrinsic value variance is analytically sound and directionally important. It correctly highlights interest rate risk as the dominant single threat to the bull case.
Special Dividend Anticipation	Gemini's projection of a $12–$15 special dividend in late 2026 or 2027 is directionally plausible given the $14.76B in operating cash flow and the historical 2–3-year cadence. This passed our analytical scrutiny.

8.2 What AI Got Wrong or Missed

Category	Detail
Intrinsic Value Inconsistency	Gemini's DCF cited an intrinsic value of $846.25/share, while the Monte Carlo output asserted a 70% probability of fair value between $920–$1,100. These two outputs are internally inconsistent, Gemini failed to reconcile the discrepancy or acknowledge the contradiction. A rigorous analyst would flag and resolve this.
AI-Driven Throughput Claim	Gemini stated that 'AI-driven checkout technology increased warehouse throughput by ~20%' in 2025/2026. This figure does not appear in Costco's SEC filings, earnings call transcripts, or any verifiable source reviewed by our team. This is a probable hallucination, a fabricated data point presented with false precision.

E-Commerce Growth Overstated	Gemini cited online sales growth of ~22% in 'the last year.' Actual SEC-verified data from Q2 FY2025 10-Q shows digital comparable sales of +20.9% (Q2) and +15.6% (full FY2025). The 22% figure conflates a single-quarter high with a full-year trend, a material misrepresentation of the true growth trajectory.
No Balance Sheet Analysis	Despite the rubric's explicit requirement for balance sheet quality assessment (tangible vs. intangible assets, goodwill structure), Gemini provided no balance sheet analysis whatsoever. This is a significant omission reflecting AI's tendency to prioritize narrative over structural accounting discipline.
Institutional Context Missing	Gemini's strategic recommendations ('HOLD for institutional portfolios; ACCUMULATE ON DIPS for growth investors') lacked any discussion of position sizing, portfolio construction context, or client suitability, standard components of a professional investment recommendation.

8.3 What We Accepted from the AI Analysis

- Membership moat framing and renewal rate emphasis, directionally correct, consistent with our human-led analysis, and useful for structuring the competitive advantage narrative.
- The identification of tariff uncertainty, geopolitical supply chain exposure, and Amazon rivalry as key risk factors, all validated against our independent research.
- The Monte Carlo's WACC variance attribution (66.7%), methodologically sound and retained as a key risk communication tool in our valuation section.
- The special dividend projection, supported by operating cash flow data from SEC filings and consistent with Costco's historical dividend cadence.

8.4 What We Discarded and Why

- The $846.25 DCF intrinsic value: Discarded because Gemini did not disclose the specific FCF projection schedule, the terminal year free cash flow, or the share count used, making the output unauditable. Our own model uses a disclosed $7.84B base FCF, ~444M shares, explicit FCF growth assumptions, and a sensitivity matrix.
- The ~20% AI-driven throughput improvement claim: Discarded entirely as an unverified, likely hallucinated figure with no SEC filing or credible press source to substantiate it. Accepting it would have introduced material misinformation into our analysis.
- The 70% probability ($920–$1,100 range) from the Monte Carlo without reconciliation to the DCF: Discarded pending our own reconciliation. Our analysis finds the Monte Carlo range plausible only under the bull-case scenario, which requires assumptions above historical norms.

8.5 How AI Enhances Equity Research, Reflections on Integration

Our experience using Gemini as an analytical assistant for this project produced several actionable conclusions about the appropriate role of AI in professional equity research:

Dimension	Our Assessment
Speed vs. Judgment	AI dramatically accelerated the synthesis of qualitative narrative, competitive moat framing, industry trend identification, and risk enumeration were produced in minutes. However, AI exercised no judgment about which risks are material, quantitatively significant, or firm specific. Speed without judgment produces comprehensive but undiscriminating output.
Pattern Recognition vs. Economic Reasoning	The Monte Carlo simulation efficiently generated 10,000 valuation scenarios and identified WACC as the dominant variance driver, a genuine contribution. But AI could not explain why WACC might change (monetary policy, credit spreads, equity risk premium shifts) or link it to macroeconomic context. Pattern recognition without causal reasoning produces insights that are difficult to act on.
Automation vs. Accountability	Gemini presented an unverified throughput figure (20% improvement from AI checkout) as fact. A human analyst is accountable for every claim in a research report. AI has no such accountability constraint; it will produce plausible-sounding numbers

	without distinguishing verified data from generated approximations. This is the most significant risk of uncritical AI adoption in financial analysis.
Future Integration Workflow	We would use AI in future workflows for: (1) first-pass competitive landscape and industry trend synthesis; (2) scenario generation and sensitivity framing; (3) earnings call summarization. We would not use AI for: primary data verification, balance sheet structure analysis, or final investment recommendation, all of which require contextual judgment, accounting literacy, and professional accountability that current AI models cannot provide.

9. Investment Recommendation & Conclusion

INVESTMENT RATING: HOLD / ACCUMULATE ON WEAKNESS

Price Target, Base Case: $850–$950 | Bear Case: $650–$750 | Bull Case: $1,100–$1,200

Costco Wholesale is one of the highest-quality retail businesses in the world. The membership flywheel, Kirkland Signature private label moat, disciplined low-SKU purchasing strategy, and 37.8% ROIC are not in question. What is in question is whether the current market price, requiring bull-case FCF growth and WACC assumptions to justify and reflects adequate margin of safety for new investors.

Our traditional analysis yields a base-case intrinsic value of $850–$950, approximately 3–13% below the current price of ~$983. Every valuation multiple sit 31–88% above its 10-year historical average. The Monte Carlo simulation identifies a meaningful probability of overvaluation at current levels, with WACC shocks and tariff escalation as the primary tail risks. We therefore recommend: (i) existing shareholders HOLD, capturing the special dividend cycle and ongoing earnings compounding; (ii) new investors ACCUMULATE ON WEAKNESS below $900, where the margin of safety becomes more defensible; and (iii) all investors monitor the membership renewal rate quarterly, as any sustained decline below 90% would represent a fundamental reassessment trigger.

AI contributed meaningfully to this analysis, particularly in risk enumeration, scenario framing, and simulation design, but required substantial human validation, correction, and supplementation. The final recommendation is grounded in SEC-verified financial data, independent DCF modeling, and professional judgment that AI tools cannot substitute for.

EXHIBITS

Exhibit A: Monte Carlo Simulation Parameters

Parameter	Specification
Number of Iterations	10,000
FCF Growth Rate	μ = 10.0%, σ = 3.0% | Normal distribution | Range: ~1%–19%
WACC (Discount Rate)	μ = 9.0%, σ = 1.2% | Normal distribution | Range: ~5.4%–12.6%
Terminal Growth Rate	μ = 3.0%, σ = 0.5% | Constrained: Terminal growth constrained to remain below WACC in all simulations
Explicit FCF Period	5 years, followed by Gordon Growth terminal value
Base FCF	$7.84B (FY2025 reported)
Shares Outstanding	~444M
Margin of Safety Applied	10% applied to all outputs
Variance Attribution	WACC: 66.7% | FCF growth: 24.0% | Terminal growth: 9.3%

Exhibit B: Sources & Data Provenance

- Costco Wholesale Corporation Form 10-K FY2025 (SEC EDGAR)
- Costco Q2 FY2025 Form 10-Q (February 2025, SEC EDGAR)
- Costco Investor Relations, IR.Costco.com
- Macrotrends.net, Historical financial data (FY2020–FY2025)
- StockAnalysis.com, Normalized income statement and FCF data
- GuruFocus.com, Historical valuation multiples and ROIC
- TipRanks / Yahoo Finance, Analyst consensus price targets
- Gemini (Google), AI-generated equity research narrative (for critical evaluation)
- Monte Carlo DCF model using a 10,000-iteration simulation implemented in @RISK (Excel add-in)

All financial figures are based on publicly available SEC filings. All valuation analyses are for academic and educational purposes only and do not constitute investment advice.